



MID Exam

Course Title	STATISTICS AND PROBABILITIES FOR ENGINEERING		Course Code	BSE0202	
Semester	Second		Academic Year	2025/2026	
Exam Date	19/04/2026	Day	Sunday	Time	2:30-3:20

Student Name		Student ID No.		Section	
--------------	--	----------------	--	---------	--

GENERAL INSTRUCTIONS

- Students are not allowed to leave the exam room during the first 30 minutes of the exam.
- Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
- Students must use a blue or black pen. Pencils are not allowed.
- Consumption of any food or beverage is prohibited except in medical cases.
- All electronic devices (e.g., cell phones) are prohibited.
- All cell phones must be turned off and placed on the invigilator's desk in the exam room.
- All notes and reading materials must be placed on the invigilator's desk in the exam room.
- Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
- Write the full information required on this page before commencing your exam.
- Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
- Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. Student's signature: _____

Question No.	Question Type	CLOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CLO, 1,8	10 (33%)			
2	Essay	CLO, 2-7	20 (67%)			
Total			30(100%)			

Moderation Review	
Comments	
Marks Reviewed by	Signature & Date

Question Number	Q1	Q2	Q3	Q4	Q5
Answer	B	B	✓	A	C

Q1: Let X be a random variable with $EX = 1$ and zero variance, then $E(2X^2 + X) =$
 A) 2 B) 3 C) 9 D) 4 E) None

MID Exam

Course Title	STATISTICS AND PROBABILITIES FOR ENGINEERING			Course Code	BSE0202
Semester	Second			Academic Year	2025/2026
Exam Date	19/04/2026	Day	Sunday	Time	2:30-3:20

Student Name		Student ID No.		Section	
--------------	--	----------------	--	---------	--

GENERAL INSTRUCTIONS

- Students are not allowed to leave the exam room during the first 30 minutes of the exam.
- Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
- Students must use a blue or black pen. Pencils are not allowed.
- Consumption of any food or beverage is prohibited except in medical cases.
- All electronic devices (e.g., cell phones) are prohibited.
- All cell phones must be turned off and placed on the invigilator's desk in the exam room.
- All notes and reading materials must be placed on the invigilator's desk in the exam room.
- Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
- Write the full information required on this page before commencing your exam.
- Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
- Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. Student's signature: _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CILO. 1,8	10 (33%)			
2	Essay	CILO. 2-7	20 (67%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by			Signature & Date

Question Number	Q1	Q2	Q3	Q4	Q5
Answer	D	D	✓	B	D

Q1: Let X be a random variable with $EX = 1$ and zero variance, then $E(3X^2 + X) =$
 A) 2 B) 3 C) 9 D) 4 E) None



Q2: For $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$, if $f(0,0) = 0.2, f(0,1) = 0.1, f(1,0) = 0.3$ and $f(1,1) = 0.4$. Then $E(XY) =$

- A) 1 B) 0.4 C) 0.8 D) 0.3 E) None

Q3: For random variables X, Y . If $\text{Corr}(x, y) = 2$ and $\sigma_X \sigma_Y = 2$. Then $\text{Cov}(2x - 3, 1 - 3y) =$

- A) 24 B) -24 C) -36 D) 36 E) none

Q4: Let X, Y be two independent random variables with $E(X) = E(Y) = 2$. Then $E(XY) =$

- A) 4 B) 9 C) 1 D) 16 E) None

Q5: If $P(A) = 0.4$ and $P(A | B) = P(B | A)$, then $P(B) =$

- A) 0.1 B) 0.2 C) 0.4 D) 0.6 E) None

Question 6: Complete the following probability distribution to have two independent random variables. (5 marks)

X \ Y	0	1	$f_X(x)$
1	0.28	0.42	0.7
2	0.12	0.18	0.3
$f_Y(y)$	0.4	0.6	1

1 true $\rightarrow 2$
 2 " $\rightarrow 3$
 3 " $\rightarrow 4$
 4 # $\rightarrow 5$



Question 7: For a random variable X with CDF $F(x) = 1 - e^{-2x}$, $x > 0$, find the pdf $f(x)$ and $P(X = 0.1)$ (5 marks)

$$f(x) = F'(x) \\ = 2e^{-2x}, \quad x > 0 \quad (3)$$

$$P(0.1) = 0 \quad (2)$$

Question 8: Consider the joint pdf $f(x, y) = 1$ when $0 < x < 1$, $0 < y < 1$ and zero otherwise. Find $P(-1 < X < 1/2 \text{ and } -1 < Y < 1/2)$. (5 marks)

$$P(0 < x < 1, 0 < y < 1) = \int_{-1}^{1/2} \int_{-1}^{1/2} 1 \, dy \, dx \quad (1)$$

$$= \int_0^{1/2} \int_0^{1/2} 1 \, dx \quad (2)$$

$$= \left(\frac{1}{2} \times \frac{1}{2}\right)$$

$$= \frac{1}{4} \quad (2)$$



Question 9: Suppose that $P(A) = 0.4$, $P(B) = 0.6$ and $P(\overline{A \cup B}) = 0.24$. Show that A and B are independent? (5 marks)

$$P(\overline{A \cup B}) = 0.24 \rightarrow P(A \cup B) = 1 - 0.24$$

$$\therefore P(A \cup B) = 0.76$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.76 = 0.4 + 0.6 - P(A \cap B)$$

$$\therefore P(A \cap B) = 1 - 0.76$$

$$= 0.24$$

$$P(A)P(B) = (0.4)(0.6) = 0.24$$

$$\therefore P(A \cap B) = P(A)P(B)$$

$\therefore A$ and B are independent

OR

$$P(\overline{A \cup B}) = P(\bar{A} \cap \bar{B})$$

$$\therefore P(\bar{A} \cap \bar{B}) = 0.24$$

$$P(\bar{A})P(\bar{B}) = (0.6)(0.4)$$

$$= 0.24$$

$$\therefore P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$$

$\therefore \bar{A}$ and $\bar{B}$ are independent

$\therefore A$ and B are independent
(Thm)